Problem Solving with Impulse and Momentum

Answer key

Conventions used throughout

• **Symbols (from the lessons):** $\vec{p} = m\vec{v}$ is momentum (kg·m/s). A prime (′) means *after* the collision; no prime means before. Subscripts name the object.

• **Method (from the lesson):** draw a before and after diagram, choose the + direction, then write $\vec{p}_{tot} = \vec{p}_{tot}'$, that is $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$.

• **Signs:** in one dimension every velocity carries its sign in the working ($v = -3.0$ m/s). The direction is written in square brackets only in the final answer.

• **Frames of reference (from the lesson):** when applying conservation of momentum, all velocities must be in the same frame of reference. A velocity given relative to a moving object is changed first: $v_{bw} = v_{br} + v_{rw}$.

• **Which law, when:** momentum is conserved during a collision (no net outside force on the system). Mechanical energy is conserved only when no energy is turned into heat, sound or permanent deformation. Energy symbols: $E_k = \frac{1}{2}mv^2$, $E_g = mgh$, $E_e = \frac{1}{2}kx^2$.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• $g = 9.81$ m/s^2. Every step keeps full precision; only final answers are rounded.

MOMENTUM AND RELATIVE MOTION

Boy walking on a raft

A 35.0 kg boy is standing on a raft that has a mass of 70.0 kg. The raft, originally at rest, is free to slide on water. The boy begins to walk along the raft at a constant speed of 2.0 m/s relative to the raft. What is the raft's velocity with respect to the water?

Hints: write an expression for the velocity of the boy with respect to the water (v_{bw} = ___ + ___)

Now use conservation of momentum with all velocities with respect to the water

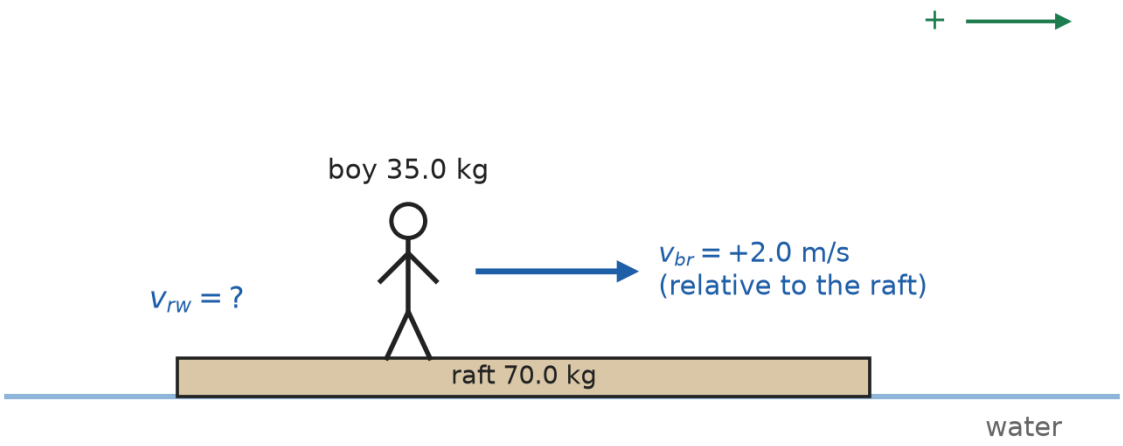
GIVEN

$m_b = 35.0$ kg $\qquad m_r = 70.0$ kg

$v_{br} = +2.0$ m/s $\qquad v_{rw} = ?$

Sign convention: the direction the boy walks is +. Subscripts: br = boy relative to raft, bw = boy relative to water, rw = raft relative to water.

The boy and the raft (+ is the way he walks)



Step 1 – The boy's velocity relative to the water (the hint)

His velocity relative to the water is his velocity relative to the raft plus the raft's velocity relative to the water.

Relative velocity: $v_{bw} = v_{br} + v_{rw}$

Step 2 – Conservation of momentum, relative to the water

Everything starts at rest, so the total momentum is zero and stays zero.

Formula: $0 = m_b v_{bw} + m_r v_{rw}$

Put in Step 1: $0 = m_b(v_{br} + v_{rw}) + m_r v_{rw}$

Expand and collect v_{rw}: $0 = m_b v_{br} + (m_b + m_r)\, v_{rw}$

Isolate v_{rw}: $v_{rw} = -\dfrac{m_b v_{br}}{m_b + m_r}$

Put in the numbers:

$$v_{rw} = -\frac{(35.0)(2.0)}{35.0 + 70.0} = -\frac{70.0}{105.0} = -0.6667 \text{ m/s}$$

The negative sign means the raft moves opposite to the way the boy walks.

FINAL ANSWER

$v_{rw} = -0.67$ m/s

$\vec{v}_{rw} = 0.67$ m/s [opposite to the boy's walking direction]

Check: the boy's velocity relative to the water is 2.0 + (−0.6667) = +1.3333 m/s, and (35.0)(1.3333) + (70.0)(−0.6667) = 0.

WHEN ENERGY AND MOMENTUM MUST BOTH BE USED

Ballistic pendulum

The following problem has two parts to it: when the bullet hits the piece of wood and then when the wood swings up to its maximum height.

a) When the bullet hits the wood, will momentum be conserved? What about energy? Why?

b) As the wood (with bullet embedded) swings upwards, is momentum be conserved? What about energy? Why?

As seen in the diagram below, a ballistic pendulum involves firing a bullet into a block of wood and using the motion of the block to determine the initial speed of the bullet. A 30.0 g bullet is fired at a 1.2 kg block hanging from a string of length 1.5 m. At the maximum height of the block, the string creates an angle of 67.6^{o} with the vertical. What was the initial speed of the bullet?

a) When the bullet hits the wood, will momentum be conserved? What about energy? Why?

Momentum is conserved. The collision is over so quickly that the only important forces are the equal and opposite forces between the bullet and the block, so the total momentum of bullet + block does not change. Kinetic energy is **not** conserved: the bullet sticks in the wood (a totally inelastic collision), and most of its kinetic energy becomes heat, sound and damage to the wood.

b) As the wood (with bullet embedded) swings upwards, is momentum be conserved? What about energy? Why?

Momentum is **not** conserved: gravity and the string's tension are outside forces that slow the block to a stop. Mechanical energy is conserved: there is no friction, the tension is always perpendicular to the motion so it does no work, and gravity just turns kinetic energy into gravitational potential energy.

Calculation What was the initial speed of the bullet?

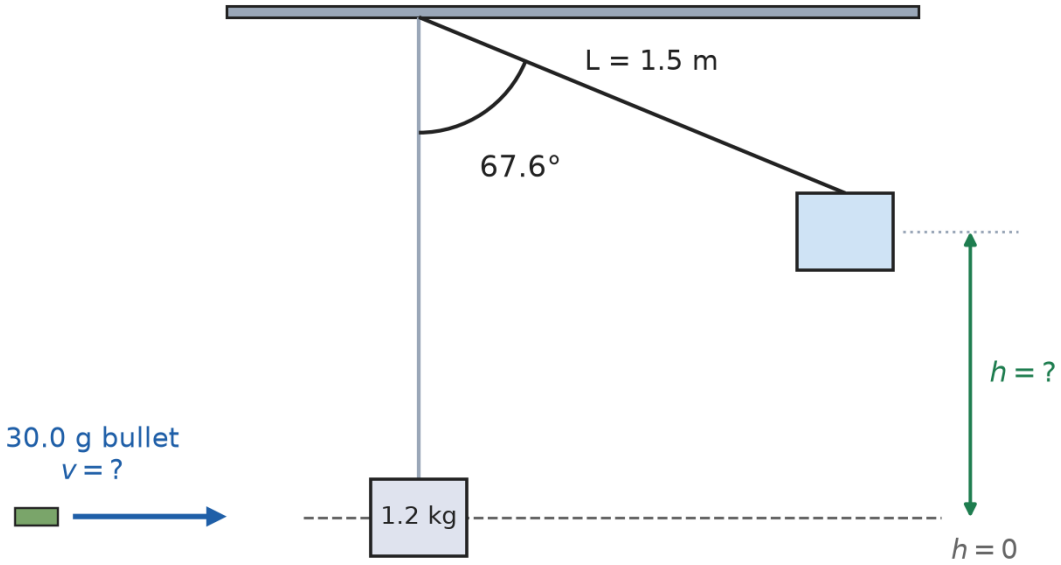
GIVEN

$m = 30.0\ \text{g} = 0.0300\ \text{kg}$ $M = 1.2\ \text{kg}$

$L = 1.5\ \text{m}$ $\theta = 67.6°$

$v = ?$

The pendulum before the bullet hits and at its highest point



Work backward: the swing (energy) gives the speed just after the collision, and the collision (momentum) then gives the bullet's speed.

Step 1 – Height gained by the block

At the highest point the block is L cos θ below the pivot, measured straight down. It started L below the pivot.

Formula: $h = L - L\cos\theta$

Put in the numbers:

$$h = 1.5 - (1.5)\cos 67.6° = 1.5 - 0.5716 = 0.9284\ \text{m}$$

Step 2 – Speed just after the collision (energy during the swing)

Bottom: all kinetic. Highest point: at rest, all gravitational. The total mass cancels.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}(m+M)v'^{2} = (m+M)gh$

Isolate v': $v' = \sqrt{2gh} = \sqrt{2(9.81)(0.9284)} = 4.2679$ m/s

Step 3 – Speed of the bullet (momentum during the collision)

The block starts at rest and the bullet stays in it, so they share the velocity v′ afterward.

Formula: $mv + M(0) = (m+M)\,v'$

Divide both sides by m: $v = \frac{(m+M)\,v'}{m}$

Put in the numbers: $v = \frac{(0.0300+1.2)(4.2679)}{0.0300} = 174.98$ m/s

FINAL ANSWER

a) Momentum: conserved. Kinetic energy: not conserved (the bullet sticks).

b) Momentum: not conserved (gravity and tension act). Mechanical energy: conserved.

Initial speed of the bullet: v = 1.7 × 10^2 m/s

Worksheet answer: 175 m/s
Why the answers differ: Same value, different rounding. The key gets 174.98 m/s. The block's mass (1.2 kg) and the string length (1.5 m) have 2 significant figures, so the key gives 1.7 × 10^2 m/s; the worksheet keeps 3.

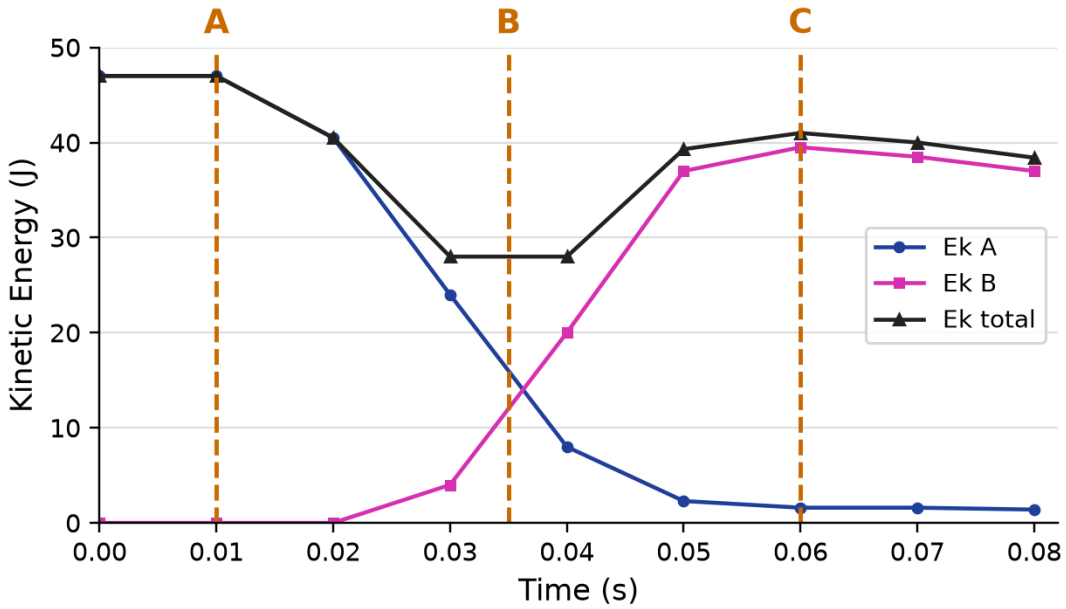
THE PROCESS OF A COLLISION

Kinetic energy during a collision

Collisions often occur very quickly so we don't usually notice what is actually happening during a collision. In this example, cart A collided with a smaller cart B using an uncompressed spring.

1. Draw a vertical line on the graph labelled "A" to indicate the moment in time when the collision begins and one labelled "C" to indicate when the collision ends.
2. What would we observe about the spring at moments "A" and "C"?
3. Why does the total kinetic energy dip down in the middle of (during) the collision? Where did it go?
4. Draw a vertical line on each graph labelled "B" to indicate the moment in time when the spring was at its maximum compression.
5. How do the speeds of the two carts compare at maximum compression?

The worksheet's graph (values read off it, approximately) with lines A, B and C



1. Draw a vertical line on the graph labelled “A” to indicate the moment in time when the collision begins and one labelled “C” to indicate when the collision ends.

A is at about 0.01 s: up to then E_k A is steady, and after it E_k A starts to fall. C is at about 0.06 s: from then on both carts’ kinetic energies have levelled off, so the carts are no longer pushing on each other.

2. What would we observe about the spring at moments “A” and “C”?

At A the spring has just touched cart B and is about to start compressing. At C it has just returned to its full length and is about to lose contact. At both moments the spring is uncompressed, so it stores no energy.

3. Why does the total kinetic energy dip down in the middle of (during) the collision? Where did it go?

During the collision the spring is compressed, and some of the carts’ kinetic energy is stored in it as elastic potential energy. As the spring pushes the carts apart it gives that energy back, so the total kinetic energy rises again.

4. Draw a vertical line on each graph labelled “B” to indicate the moment in time when the spring was at its maximum compression.

B is at about 0.035 s, where the total kinetic energy is lowest. The least kinetic energy means the most energy stored in the spring, which is the greatest compression.

5. How do the speeds of the two carts compare at maximum compression?

They are equal: at maximum compression both carts have the same velocity. Before that moment cart A is faster than cart B, so the gap is closing and the spring is still being squeezed. After it cart B is faster, so the gap opens. (Cart A still has more kinetic energy at B because it has more mass.)

FINAL ANSWER

1. A ≈ 0.01 s (collision begins); C ≈ 0.06 s (collision ends)

2. The spring is uncompressed at both A and C (just touching, then just letting go)

3. The missing kinetic energy is stored in the compressed spring as elastic potential energy

4. B ≈ 0.035 s, at the lowest point of the total kinetic energy

5. The two carts have the same speed (the same velocity)

PRACTICE

Two carts cushioned by a spring

Cart 1 has mass of 1.8 kg and is moving with a velocity of 4.0 m/s [right] along a frictionless track. Cart 2 has a mass of 2.2 kg and is moving at 6.0 m/s [left]. The carts collide in a head-on elastic collision cushioned by a spring with a spring constant of 8.0 x 10^4 N/m.

a) Determine the compression of the spring during the collision at the moment when Cart 2 is moving at 4.0 m/s [left].

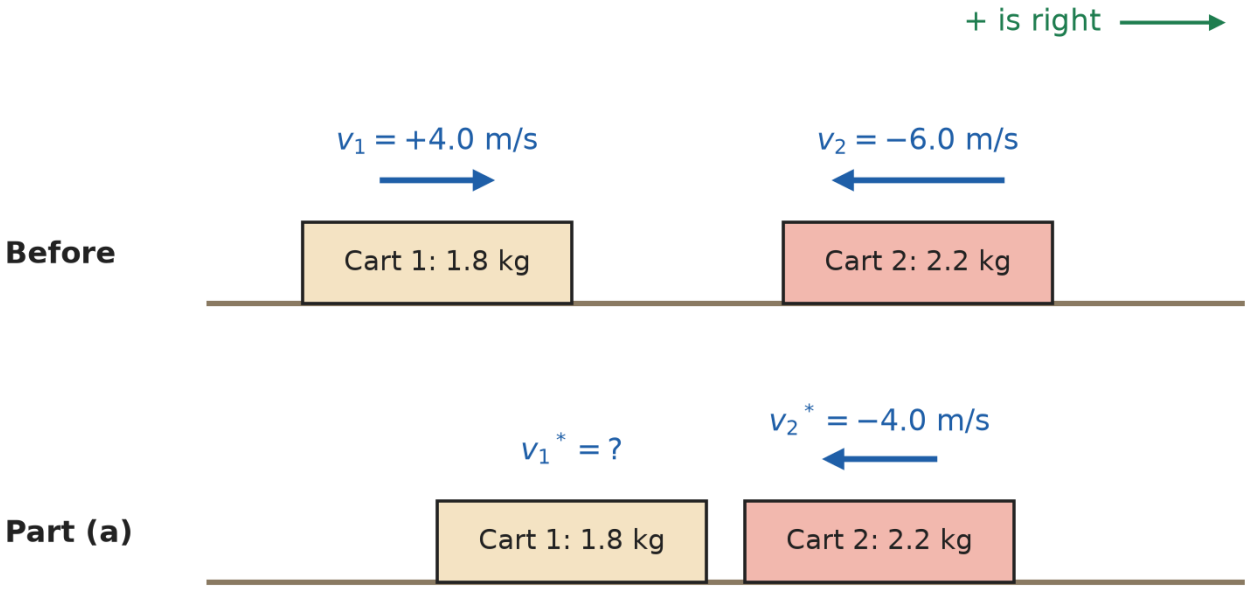
b) Determine the maximum compression of the spring.

GIVEN

$m_1 = 1.8$ kg	$v_1 = +4.0$ m/s	$m_2 = 2.2$ kg
$v_2 = -6.0$ m/s	$k = 8.0 \times 10^4$ N/m	$x = ?$

Sign convention: right is +. A star (*) marks a velocity during the collision.

Before the collision, and at the moment in part (a)



During the collision the only horizontal forces are the spring forces the carts put on each other, so momentum is conserved at every moment. The collision is elastic and the track is frictionless, so the total energy (kinetic + elastic) is also conserved at every moment.

Total kinetic energy before the collision (used in both parts)

Formula: $E_{TOTi} = \frac{1}{2}m_1 v_1^2 + \frac{1}{2}m_2 v_2^2$

Put in the numbers:

$$E_{TOTi} = \frac{1}{2}(1.8)(4.0)^2 + \frac{1}{2}(2.2)(-6.0)^2 = 14.4 + 39.6 = 54.0 \text{ J}$$

a) Determine the compression of the spring during the collision at the moment when Cart 2 is moving at 4.0 m/s [left].

Step 1 – Velocity of cart 1 at that moment (momentum)

Formula: $m_1v_1 + m_2v_2 = m_1{v_1}^* + m_2{v_2}^*$

Isolate ${v_1}^*$: $${v_1}^* = \frac{m_1v_1 + m_2v_2 - m_2{v_2}^*}{m_1}$$

Put in the numbers:

$${v_1}^* = \frac{(1.8)(4.0) + (2.2)(-6.0) - (2.2)(-4.0)}{1.8} = \frac{7.2 - 13.2 + 8.8}{1.8}$$

${v_1}^* = +1.5556$ m/s

Step 2 – Compression of the spring (energy)

The kinetic energy that is missing at this moment is stored in the spring.

Formula: $$E_{TOTi} = \frac{1}{2}m_1{v_1}^{*2} + \frac{1}{2}m_2{v_2}^{*2} + \frac{1}{2}kx^2$$

Kinetic energy now:

$$\frac{1}{2}(1.8)(1.5556)^2 + \frac{1}{2}(2.2)(-4.0)^2 = 2.1778 + 17.6 = 19.7778 \text{ J}$$

Energy in the spring: $$\frac{1}{2}kx^2 = 54.0 - 19.7778 = 34.2222 \text{ J}$$

Isolate x: $$x = \sqrt{\frac{2(34.2222)}{8.0 \times 10^4}} = 0.02925 \text{ m}$$

b) Determine the maximum compression of the spring.

Step 1 – Shared velocity at maximum compression (momentum)

At maximum compression the two carts have the same velocity v* (see “The process of a collision”, question 5).

Formula: $m_1 v_1 + m_2 v_2 = (m_1 + m_2)\, v^*$

Isolate v^*: $v^* = \dfrac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \dfrac{7.2 - 13.2}{1.8 + 2.2} = -1.50 \text{ m/s}$

Step 2 – Maximum compression (energy)

Kinetic energy now: $\dfrac{1}{2}(m_1 + m_2)v^{*2} = \dfrac{1}{2}(4.0)(-1.50)^2 = 4.50 \text{ J}$

Energy in the spring: $\dfrac{1}{2}kx^2 = 54.0 - 4.50 = 49.50 \text{ J}$

Isolate x: $x = \sqrt{\dfrac{2(49.50)}{8.0 \times 10^4}} = 0.03518 \text{ m}$

FINAL ANSWER

a) x = 0.029 m = 2.9 cm

b) x = 0.035 m = 3.5 cm

SIN QUESTION 1

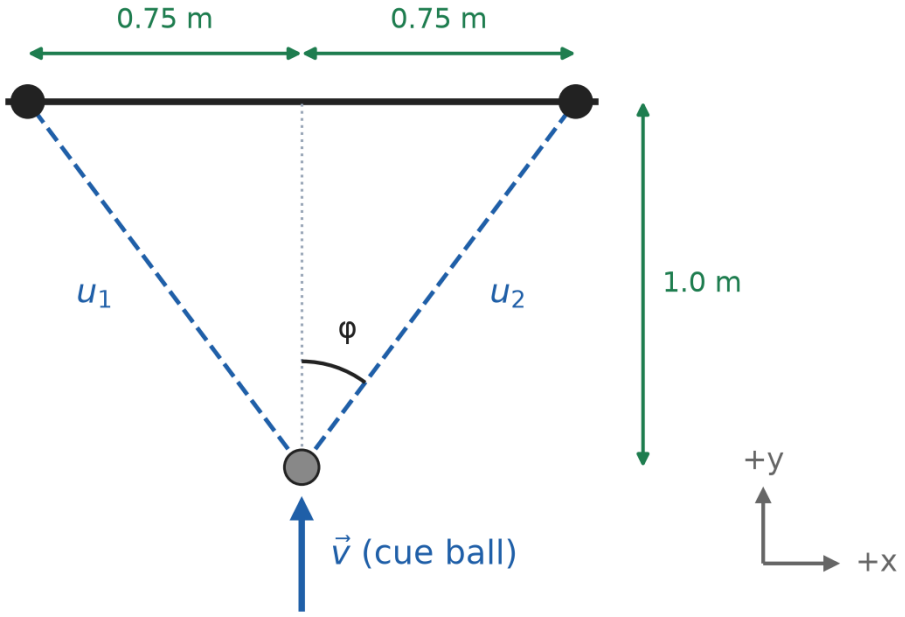
Willie’s pool shot

The short poem printed above this question on the worksheet is not copied here; it has no data.

One of Willie’s worst shots is shown below. He sends his white cue ball with velocity v so that it strikes a coloured ball at rest. To his dismay, both balls end up in different pockets, as shown! Using the given geometry, calculate the fraction of kinetic energy lost in the collision. Treat the balls as identical point masses and neglect any friction or rotational motion.

a) 0 b) 1/7 c) 1/5 d) 7/32 e) 4/9

The geometry from the worksheet (+y is the cue ball’s original direction)



Each ball goes 0.75 m sideways and 1.0 m forward to reach its pocket.

Step 1 – Direction of each ball after the collision

Each path is the hypotenuse of a right triangle with sides 0.75 m and 1.0 m. φ is the angle between a path and the y-direction.

Path length: $\sqrt{(0.75)^2 + (1.0)^2} = 1.25 \text{ m}$

Angle: $\cos\varphi = \dfrac{1.0}{1.25} = 0.80$ $\sin\varphi = \dfrac{0.75}{1.25} = 0.60$

Step 2 – Conservation of momentum in x and y

The balls have the same mass m. Before the collision there is no x-momentum.

x: $0 = mu_1 \sin\varphi - mu_2 \sin\varphi \quad \Rightarrow \quad u_1 = u_2 = u$

y: $mv = mu\cos\varphi + mu\cos\varphi = 2mu\cos\varphi$

Isolate u: $u = \dfrac{v}{2\cos\varphi} = \dfrac{v}{2(0.80)} = 0.625v$

Step 3 – Fraction of kinetic energy lost

Before: $E_k = \dfrac{1}{2}mv^2$

After: $E_k' = 2\left(\dfrac{1}{2}mu^2\right) = m(0.625v)^2 = 0.390625\,mv^2$

Fraction kept: $\dfrac{E_k'}{E_k} = \dfrac{0.390625\,mv^2}{0.5\,mv^2} = 0.78125 = \dfrac{25}{32}$

Fraction lost: $1 - \dfrac{25}{32} = \dfrac{7}{32}$

FINAL ANSWER

d) 7/32 of the kinetic energy is lost

SIN QUESTION 2

Millie and Pam on skates

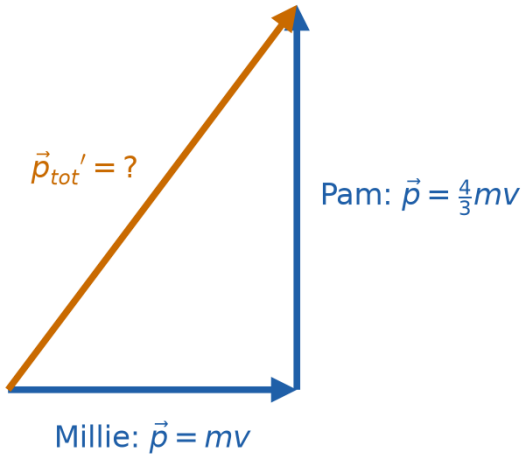
The short poem printed above this question on the worksheet is not copied here. It supplies the set-up: Millie collides with Pam, they hold on tight and slide off as one (a totally inelastic collision).

If their initial velocities along the horizontal frictionless ice surface are equal in magnitude but at right angles, and if Pam is more massive than Little Millie by a factor of 4/3, find the ratio of their final to initial speeds.

A) 3/4 B) 4/5 C) 5/7 D) 9/16 E) 2/3

Call Millie's mass m and the initial speed of each skater v. Pam's mass is (4/3)m. Take Millie moving along +x and Pam along +y. They stick together, so momentum is conserved in x and in y.

The two momentum vectors, added tip to tail



Step 1 – Total momentum

The two momenta are perpendicular, so use the Pythagorean theorem.

x: $p'_x = mv$

y: $p'_y = \frac{4}{3}mv$

Size: $$p'_{tot} = \sqrt{(mv)^2 + \left(\frac{4}{3}mv\right)^2} = mv\sqrt{1 + \frac{16}{9}} = mv\sqrt{\frac{25}{9}} = \frac{5}{3}mv$$

Step 2 – Final speed

Together their mass is m + (4/3)m = (7/3)m.

Formula: $$v' = \frac{p'_{tot}}{m_{tot}} = \frac{\frac{5}{3}mv}{\frac{7}{3}m} = \frac{5}{7}v$$

Ratio: $$\frac{v'}{v} = \frac{5}{7}$$

FINAL ANSWER

C) 5/7

SIN QUESTION 3

Shooting Turbot to get over the hill

Fisheries minister Brian Tobin dreamed of a game he played in his youth. He was coasting over a level frictionless ice surface riding on a 10 kg sled loaded with himself (mass 50 kg) and 5 identical frozen Turbot (mass 2 kg each fish). He was being pursued by the Spanish fleet armed with fine mesh nets. His speed was 10 m/s, and he had to get over a 6 m high hill to escape. He carried a sling shot, registered with justice minister Allan Rock, and could shoot the Turbot backwards with a horizontal speed of 20 m/s relative to his sled. Assuming he shot one fish at a time, and finished all his shots before he reached the hill, calculate the minimum number of Turbot he had to shoot in order to escape over the hill.

A) None B) One C) Two D) Three E) Four

GIVEN

$m_{sled} = 10$ kg	$m_{Tobin} = 50$ kg	$m_{fish} = 2$ kg each
$v = +10$ m/s	$h = 6$ m	$v_{fs} = -20$ m/s

Sign convention: the direction he is coasting is +. v_{fs} is the fish's velocity relative to the sled.

Step 1 – Speed needed at the bottom of the hill (energy)

The ice is frictionless. To just reach the top, all of the kinetic energy at the bottom becomes gravitational energy. The mass cancels.

Formula: $\frac{1}{2}mv_{min}^2 = mgh$

Isolate v_{min}: $v_{min} = \sqrt{2gh} = \sqrt{2(9.81)(6)} = 10.850$ m/s

He is only going 10 m/s, so without shooting he does not make it. Each fish shot backward pushes the sled forward.

Step 2 – What one shot does (momentum, relative to the ice)

Call the total mass before a shot M and the sled's velocity before and after it v and v′. The fish leaves at −20 m/s relative to the sled, so its velocity relative to the ice is v′ − 20.

Formula: $Mv = (M-2)\,v' + (2)(v' - 20)$

Expand: $Mv = Mv' - 40$

Isolate v': $v' = v + \frac{40}{M}$

Step 3 – Shoot until the speed is high enough

At the start M = 10 + 50 + 5(2) = 70 kg. After each shot M is 2 kg less.

Shot 1 (M = 70 kg):

$$v' = 10 + \frac{40}{70} = 10.571 \text{ m/s} \quad \text{(less than 10.850)}$$

Shot 2 (M = 68 kg):

$$v' = 10.571 + \frac{40}{68} = 11.160 \text{ m/s} \quad \text{(more than 10.850)}$$

FINAL ANSWER

C) Two